

Assignment 2 — EDA & Data Visualization

Preparing a real dataset for machine learning

Modules covered: Exploratory Data Analysis (Modules 1–7) · Data Visualization (Modules 1–7)

Dataset: House Prices — Advanced Regression Techniques (Kaggle)

Weight: 100 points (+8 bonus)

Estimated effort: 8–10 hours

Submission: one PDF on the leading2ai platform (see §12)

0. The scenario

You have just joined a small real-estate analytics team in Ames, Iowa. Next week a teammate will train a model that predicts a house's **sale price**. Your job this week is everything that has to happen *before* that model: understand the data, find what is wrong with it, decide how to fix each problem, and hand over a dataset that a model can learn from safely.

You will also hand over a written record of every decision you made, so your teammate can trust your work.

You do not train a model in this assignment. That comes later. This assignment is about the preparation that decides whether the model will be any good.

1. How this assignment works

Part	Topic	What you do	Points
A	First look & data quality	Inspect the data, catch wrong types, near-constant columns and impossible values	12
B	Missing values	Tell "missing" apart from "absent", investigate why values are missing, decide what to do	14
C	Distributions, outliers, transformations	Study the target and the features, find outliers, decide whether to keep, cap, transform or remove them	14
D	Relationships & correlation	Compare groups, read a correlation heatmap, find redundant features	14
E	Visualization craft & storytelling	Plot sales over time, fix a misleading chart, tell a one-page data story	12
F	Make it ML-ready	Split, impute, engineer, encode, scale, save, plus the data-preparation log	18
G	Written questions	Eight questions about your own results and choices	16

Five rules apply to every part:

1. **Show your output.** Every task must show its result, as a printed table, a number or a chart. A task with code but no visible output earns at most half marks.

2. **Every chart passes the chart checklist** from the Data Visualization lecture (see §3). Each chart that breaks it loses marks.
3. **Evidence, not opinion.** Every written answer must cite a number, a value or a chart from *your own notebook*. An answer that would be equally true for any dataset scores zero.
4. **Your own words.** Answers copied from the slides, from Kaggle notebooks or generated by an AI tool score zero. Write as if you were explaining to a classmate who missed the session.
5. **Never edit the CSV by hand.** Every change to the data must happen in your notebook, where it can be seen and repeated. Use random seed **42** wherever randomness is involved.

2. Dataset and setup

Go to Kaggle: kaggle.com/competitions/house-prices-advanced-regression-techniques/data. Sign in, accept the competition rules and download:

- `train.csv`: the data for this whole assignment (you need `test.csv` only for the bonus).
- `data_description.txt`: the data dictionary. **Keep it open while you work.** Several of the "problems" in this dataset are explained there, and reading it is part of the assignment.

Work in one Jupyter notebook. You will need pandas, NumPy, Matplotlib and Seaborn. scikit-learn is optional in Part F; everything there can also be done with pandas.

Checkpoint: before you start, check that the raw file has **1,460 rows and 81 columns**. The target is `SalePrice`, in US dollars. `Id` is a row identifier, not a feature.

Working columns

Parts A and B use all 81 columns. From Part C onward, work with this subset of 39 columns so the charts stay readable (you may add more if you justify them):

Group	Columns
Target	<code>SalePrice</code>
Size & lot	<code>LotFrontage</code> , <code>LotArea</code> , <code>GrLivArea</code> , <code>TotalBsmtSF</code> , <code>1stFlrSF</code> , <code>2ndFlrSF</code> , <code>GarageArea</code> , <code>PoolArea</code> , <code>MasVnrArea</code>
Counts	<code>FullBath</code> , <code>HalfBath</code> , <code>BsmtFullBath</code> , <code>BsmtHalfBath</code> , <code>BedroomAbvGr</code> , <code>TotRmsAbvGrd</code> , <code>Fireplaces</code> , <code>GarageCars</code>
Time	<code>YearBuilt</code> , <code>YearRemodAdd</code> , <code>GarageYrBlt</code> , <code>YrSold</code> , <code>MoSold</code>
Ratings	<code>OverallQual</code> , <code>OverallCond</code> , <code>ExterQual</code> , <code>KitchenQual</code> , <code>BsmtQual</code> , <code>FireplaceQu</code>
Categories	<code>MSSubClass</code> , <code>MSZoning</code> , <code>Neighborhood</code> , <code>BldgType</code> , <code>HouseStyle</code> , <code>GarageType</code> , <code>CentralAir</code> , <code>SaleCondition</code> , <code>Utilities</code> , <code>Street</code>

3. The chart checklist (applies to every chart)

1. The **title states the finding**, not just the variables. "Sale price rises steeply with overall quality" is a finding. "SalePrice vs OverallQual" is not.
2. Both axes are labelled, **with units** (\$, square feet, years, count).

3. The chart type matches the question: comparison, distribution, relationship, trend or composition.
4. Bar charts start at zero. Categories are sorted in a meaningful order: by value, or in their natural order for ratings.
5. No 3D, no decorative colour, no pie charts with many slices. Colour is used only where it carries meaning.
6. The chart is readable when printed in your PDF. Tick labels must not overlap.
7. Below each chart, write **one sentence** saying what it shows.

4. Part A — First look & data quality (12 points)

Lecture: EDA Module 1 (Inspecting a new dataset) and Module 2 (Data quality).

A1 — The first-look checklist (3 pts)

Run the five-question first-look checklist from the lecture on the full dataset. Show: the shape; how many columns there are of each data type; the first and last five rows; summary statistics for the numeric columns; and the number of distinct values in each text column.

Write (4–6 sentences): name **three things that look fine** and **three things that raise a flag**. Back each one with a specific value from your output.

A2 — Types that lie (3 pts)

A column's stored type does not always match what the column means. Look up `MSSubClass` in `data_description.txt`. It is stored as an integer. Show its distinct values, then convert it to a categorical type and show the type before and after.

Then classify each of these as **continuous numeric**, **count**, **nominal category** or **ordinal category**, and justify each in one line: `MSSubClass`, `OverallQual`, `OverallCond`, `KitchenQual`, `MoSold`, `YrSold`, `GarageCars`.

Write (2–3 sentences): if `MSSubClass` stayed an integer, a model would treat class 60 as "twice" class 30. Explain why that is nonsense for this column, and what kind of wrong pattern a linear model could learn from it.

A3 — Duplicates and near-constant columns (3 pts)

Check for fully duplicated rows and for duplicated `Id` values, and report both counts. Then, for every column, compute the share of rows taken by its single most frequent value. List every column where that share is above 95%, as a table sorted from highest to lowest.

Checkpoint: `Utilities` (1,459 of 1,460 rows have the same value) and `Street` (1,454 of 1,460) should both be on your list.

Write (2–3 sentences): explain why a column that is almost constant gives a model almost nothing to learn from, and what can go wrong when the rare value lands only in the test set. State your keep/drop decision for `Utilities` and `Street`.

A4 — Values that are possible, but suspicious (3 pts)

Write a rule-based check for each of the following and report the number of rows that break it, in one table:

1. The house was remodelled before it was built (`YearRemodAdd` earlier than `YearBuilt`).
2. The garage was built before the house (`GarageYrBlt` earlier than `YearBuilt`).
3. The house was sold before it was built (`YrSold` earlier than `YearBuilt`).
4. A year lies in the future relative to the data (any year column later than 2010).
5. The house has a garage type but a garage area of zero, or the other way round.

Write: for each rule that finds rows, show those rows, then decide **fix automatically** or **flag for a human** (the two lists from the lecture), with a one-line reason. A garage built a few years before the house may be genuine. Say how you would tell a real case from a typo.

5. Part B — Missing values (14 points)

Lecture: EDA Module 3 (Detecting and handling missing values).

B1 — Missingness audit (3 pts)

Produce a table of every column with at least one missing value, showing the count and the percentage, sorted from most to least missing. Then draw a **horizontal bar chart** of the percentages. Add a reference line at 50%.

Checkpoint: PoolQC tops the list with 1,453 missing (99.5%). Roughly 19 columns have gaps.

B2 — Missing, or absent? (4 pts)

Read the entries for PoolQC, Alley, Fence, FireplaceQu, GarageType and BsmtQual in `data_description.txt`. Then **prove from the data** what "NA" means there. For example, show that the number of missing FireplaceQu values equals the number of houses with zero Fireplaces. Do the same for pools (against PoolArea) and garages (against GarageArea).

Now look at MasVnrType. Compare the missing count your notebook reports with the values `data_description.txt` says the column can contain. If the count looks far too high, find out why. Hint: look at which text values your CSV loader treats as missing by default. Fix it *at load time* and show the corrected count.

Write (3–4 sentences): explain why filling PoolQC with its most frequent value would be a serious error, and what the "missing" values in these columns actually are. What did the MasVnrType case teach you about trusting a missing-value count?

B3 — Truly missing: LotFrontage (4 pts)

LotFrontage (the length of street connected to the lot, in feet) is missing for about 18% of houses, and here "NA" does *not* mean "no street". Investigate **why** it is missing:

- Compare the missing rate across Neighborhood and across LotConfig with a sorted bar chart.
- Compare LotArea for rows with and without LotFrontage, using side-by-side box plots.

Then compare three ways of filling it: **overall mean**, **overall median** and **median within the same neighborhood**. Overlay the distribution of the observed values with the distribution after each kind of filling, as histograms or KDEs on one set of axes or as three small panels.

Write (3–4 sentences): is the missingness MCAR, MAR or MNAR? Point to your charts. Which filling method distorts the distribution least, and why does filling with a single value create a spike? (*Explore only. You will apply the final choice in Part F, after splitting.*)

B4 — The missing-value decision table (3 pts)

For **every** column with missing values, write one row:

Column	% missing	Kind (absent / MCAR / MAR / MNAR)	Decision	One-line reason
PoolQC	99.5	...	...	...

Possible decisions: fill with a "None" category, fill with 0, fill with a group median, fill with the mode, drop the column, or any of these **plus** a "was missing" indicator column. Two need special thought:

- GarageYrBlt for houses with no garage. There is no sensible year to fill in. What do you do, and why?
- Electrical, which has a single missing value.

6. Part C — Distributions, outliers & transformations (14 points)

Lecture: EDA Module 4 (Outliers) and Module 5 (Univariate analysis). Data Visualization Module 4 (Distributions).

C1 — Know your target (4 pts)

Plot `SalePrice` as a histogram with a KDE and as a box plot, side by side as subplots. Report the mean, median, minimum, maximum and skewness. Then apply a **$\log(1 + x)$** transform and repeat the plots and the skewness.

Checkpoint: mean $\approx$ \$180,900, median = \$163,000, skewness $\approx$ 1.88. After the log transform the skewness should be close to 0.

Write (3–4 sentences): how did the gap between the mean and the median warn you about the shape before you plotted it? Why does a long right tail in the target matter for a model that minimises squared error? Think about which houses would dominate the error. What does predicting the log of price change about how errors on cheap and expensive houses are weighted?

C2 — Skew in the features (3 pts)

Compute the skewness of every numeric working column and show it as a sorted table. Draw a grid of histograms (subplots, one figure) for the **six most skewed** features.

Write: for each of the six, decide **log-transform**, **leave as is**, or **replace with a yes/no indicator**, with a one-line reason. Hint: `PoolArea` is 0 for 1,453 houses. Is a log transform the right tool there, or is "has a pool" more honest?

C3 — Finding outliers (4 pts)

For `GrLivArea`, `LotArea` and `SalePrice`, flag outliers two ways: the **IQR rule** ($1.5 \times \text{IQR}$ beyond $Q1$ or $Q3$) and the **z-score rule** ($|z| > 3$). Show the counts in a 3×2 table.

Then draw a scatter plot of `GrLivArea` (x) against `SalePrice` (y). Colour the houses with more than 4,000 sq ft of living area differently and **annotate** them on the chart.

Checkpoint: four houses exceed 4,000 sq ft. Two of them sold for well under \$200,000, far below what their size suggests.

Write (3–4 sentences): why do the IQR rule and the z-score rule disagree so much on skewed columns? For the two cheap, very large houses, look at their `SaleCondition` and `Neighborhood` and read what `SaleCondition` means in the data dictionary. Are they errors, or genuine but unusual sales?

C4 — Keep, cap, transform or remove? (3 pts)

Apply the lecture's four options to your findings, and **show the effect** of each choice:

- The correlation between GrLivArea and SalePrice, with and without the two cheap, very large houses.
- LotArea before and after capping at the 1st and 99th percentiles, as side-by-side box plots.

Write (3–4 sentences): state your decision for each outlier group and defend it. Removing two rows changed a correlation. What does that tell you about how much weight a few points can carry? These decisions go into your data-preparation log (F5).

7. Part D — Relationships & correlation (14 points)

Lecture: EDA Modules 5–7 (Multivariate analysis, Correlation, Feature–target relationships). Data Visualization Modules 3–4 (Seaborn, Relationships).

D1 — Categories and rare levels (3 pts)

Draw a sorted bar chart of how many houses there are per `Neighborhood`. Report how many neighborhoods have fewer than 20 sales. Show the value counts of `MSZoning` and `BldgType`.

Checkpoint: there are 25 neighborhoods. The smallest, `Blueste`, has only 2 sales.

Write (2–3 sentences): what problem does a category with 2 rows cause when you later split into train and test and one-hot encode? Propose a rule for grouping rare levels, and say which data the rule's counts should come from.

D2 — Target by group (4 pts)

1. A **horizontal box plot** of `SalePrice` by `Neighborhood`, sorted by median price.
2. A box plot of `SalePrice` by `OverallQual`.
3. A bar chart of **median** `SalePrice` by `ExterQual` and by `KitchenQual`, with the categories in their **natural order** (`Po` < `Fa` < `TA` < `Gd` < `Ex`), not alphabetical order.

Write (3–4 sentences): which of these features separates prices most clearly? Do the quality ratings step up in a steady order? Use that to argue whether they should be encoded as ordered numbers (ordinal) or as separate yes/no columns (one-hot). Why did you use the median rather than the mean in chart 3?

D3 — The correlation heatmap (4 pts)

Draw a heatmap of the Pearson correlations between all numeric working columns, including the target. Use a **diverging palette centred at zero**, annotate the cells, and hide the redundant upper triangle. Then list the **top 10 features by absolute correlation with `SalePrice`**, and next to each give its Spearman correlation too.

Checkpoint: `OverallQual` ≈ 0.79 and `GrLivArea` ≈ 0.71 should lead the Pearson list.

Write (3–4 sentences): pick one feature where Pearson and Spearman differ noticeably and explain why (think about outliers and curved relationships). Then pick one feature with a weak correlation and explain why a weak *linear* correlation does not prove the feature is useless.

D4 — Redundant features (3 pts)

List every pair of *features* (not the target) with $|r| > 0.8$. Then draw a pair plot of your five strongest predictors of price.

Checkpoint: `GarageCars` and `GarageArea` correlate at about 0.88.

Write: for each pair, explain in one line *why* the two move together (are they measuring the same physical thing?), and which one you would keep. Why is keeping both a problem for a linear model's coefficients, even if it barely changes its predictions?

8. Part E — Visualization craft & storytelling (12 points)

Lecture: Data Visualization Modules 2 (Matplotlib), 5 (Time series), 6 (Best practices) and 7 (Storytelling).

E1 — Sales over time (4 pts)

Combine `YrSold` and `MoSold` into a monthly date. Draw two line charts, stacked as subplots that share the x-axis:

1. The **number of sales per month** across the whole period.
2. The **median sale price per month**, with a 3-month rolling median overlaid in a second, clearly labelled line.

Checkpoint: sales run from January 2006 to **July 2010**. The final year is incomplete.

Write (3–4 sentences): is there a seasonal pattern in the *number* of sales? Is there one in the *price*? Can you see the 2008 financial crisis in either chart? What wrong conclusion would someone draw by comparing total sales in 2010 with 2009 without noticing the data ends in July?

E2 — Fix a misleading chart (4 pts)

First, build this deliberately **bad** chart: a bar chart of the *mean SalePrice* by `CentralAir`, with the y-axis starting at \$100,000, a different bright colour per bar, no axis labels and the title "Chart". Then build the **fixed** version next to it in the same figure.

Write: list every change you made and link each one to an item in the chart checklist (§3). By how much did the bad version exaggerate the visual gap between the two bars? Compute the ratio of bar heights as drawn against the true ratio.

E3 — A one-page data story (4 pts)

Your teammate will read this before training the model. Answer the question: "**What drives sale price in Ames, and what should the model be careful about?**"

- Use at most **three** charts. You may reuse and polish earlier ones.
- Each chart has a title that states a finding and at least **one annotation** (an arrow or a text label) pointing at the thing that matters.
- Write 150–200 words following the lecture structure: **Context** → **Insight** → **Action**. The "Action" must be concrete advice for the modeller.

9. Part F — Make it ML-ready (18 points)

Lecture: EDA Module 7 (From EDA to modeling decisions). Ties together everything above.

Until now you explored. Now you **prepare**. The order of the steps below matters, so do not rearrange them.

F1 — Split first (3 pts)

Start again from the raw file, loaded with your fix from B2. Apply only the **row-independent** fixes: type corrections (A2), filling "absent" NAs with "None" or 0 (B2), and dropping the columns you decided to drop. Then split the rows **80 / 20 into train and test with seed 42**. Print both shapes.

Checkpoint: 1,168 train rows and 292 test rows.

Write (3–4 sentences): explain **data leakage** using this dataset. If you computed the neighborhood medians for LotFrontage on all 1,460 rows before splitting, what information from the test set would leak into training, and why would your test score then be too optimistic? Why were the fixes above safe to apply before the split?

F2 — Impute and handle outliers (4 pts)

- Fill LotFrontage using the method you chose in B3. **Learn** the fill values from the training set only, then **apply** them to both sets. Handle a neighborhood that appears in test but has no training value for this column.
- Apply the rest of your B4 decision table the same way.
- Apply your outlier decisions from C4 to the **training set only**. Learn any capping limits from training data.

Checkpoint: print the total number of missing values in train and in test. Both must be **0**.

F3 — Engineer features (4 pts)

Create at least these features, in both sets:

Feature	Meaning
TotalSF	Basement + first floor + second floor area
HouseAge	Years between building and sale
YearsSinceRemodel	Years between the last remodel and the sale
TotalBathrooms	Full baths + $\frac{1}{2} \times$ half baths, above ground and in the basement
HasGarage, HasPool, HasFireplace, Has2ndFloor	Yes/no (1/0) indicators

Then invent **one feature of your own** and justify it from something you saw in Parts C–E.

Write: show a small table comparing each new feature's correlation with log sale price against the correlations of the columns it was built from. Did combining them help? Did any engineered feature make an original column redundant (see D4)?

F4 — Encode the categories (4 pts)

- **Ordinal ratings** (`ExterQual`, `KitchenQual`, `BsmtQual`, `FireplaceQu`): map them to ordered numbers, with "None" = 0 and Po = 1 up to Ex = 5.
- **Binary** (`CentralAir`): map to 1/0.
- **Rare levels**: apply your rule from D1, with the counts taken from **train**.
- **Nominal categories** (`Neighborhood`, `MSZoning`, `MSSubClass`, `BldgType`, `HouseStyle`, `GarageType`, `SaleCondition`): one-hot encode.
- Make sure the test set ends up with **exactly the same columns, in the same order**, as the train set, including when a category appears in only one of them.

Write (2–3 sentences): print the number of columns before and after encoding. Why is one-hot encoding the wrong choice for `KitchenQual`, and why is ordinal encoding the wrong choice for `Neighborhood`?

F5 — Transform, scale, save, and log your decisions (3 pts)

1. Log-transform the target and the skewed features you chose in C2.
2. Standardise the continuous numeric features (subtract the mean, divide by the standard deviation), using the **train** mean and standard deviation for both sets. Do not scale the 0/1 indicator columns or the target. Show one feature's histogram before and after.
3. Save `train_prepared.csv` and `test_prepared.csv`. Print the final shapes and confirm that every column is numeric and nothing is missing.
4. Write the **data-preparation log**: one table covering every column you changed, dropped or created.

Column	Problem found	Evidence (task)	Decision	Learned from train only?
LotFrontage	18% missing, varies by neighborhood	B3	Neighborhood median + missing flag	Yes
...	...	...	...	...

10. Part G — Written questions (16 points)

Eight questions, 2 points each. Answer each in **3–6 sentences**, and refer to your own results. A correct answer with no link to your notebook earns at most 1 point.

1. **Exploring before splitting.** In Parts A–E you explored all 1,460 rows before splitting in Part F. Is any of that exploration a form of leakage? Which of your decisions were only *informed* by the full data, and which actually *computed a number* from it that ended up in the prepared dataset?
2. **The test set.** You removed or capped outliers in the training set but not in the test set. Why? What does the test set stand for, and what would removing its outliers do to your view of how well the model will perform?
3. **Depends on the model.** Which of your preparation steps would become unnecessary if your teammate used a decision-tree-based model instead of linear regression or k-nearest neighbours? Which steps would still matter no matter what the model is? Name at least two of each.
4. **Available at prediction time?** The model is meant to price a house *when it is listed for sale*. Go through your working columns. Which would *not* be known at that moment (look closely at `SaleCondition`, `YrSold` and `MoSold`)? What is this kind of leakage called, and what should you do with those columns?
5. **Imputation assumptions.** Filling `LotFrontage` with the neighborhood median assumes something about how lots are laid out. What is the assumption, when might it fail, and what does the "was missing" indicator let the model do that the filled value alone cannot?
6. **Mean or median.** You had to choose between the mean and the median at least three times: summarising the target, comparing groups and filling missing values. Pick one of those places and explain, with numbers from your notebook, what would have gone wrong with the other choice.
7. **Charts that change decisions.** Name one chart from this assignment that changed a preparation decision you would otherwise have made, and one decision you could have made correctly from summary statistics alone. What does that say about when a chart is worth drawing?
8. **Is it ready?** Your teammate asks "can I train on this now?" Name the **two decisions in your log you are least confident about**. For each, say what evidence would change your mind and how that decision could bias the model's predictions, and for which kinds of houses.

11. Bonus (up to +8)

+5 — A reproducible preparation pipeline. Wrap Part F into functions so that a single call, taking the raw training data and returning the prepared train and test sets, reproduces your F5 files exactly from the raw CSV. Every value it learns (medians, capping limits, rare-level lists, scaling parameters) must come from the training rows only. Show that running it twice gives identical output.

+3 — Train–serve skew. Run your pipeline on Kaggle's `test.csv`, the houses a real model would have to price. Report (a) columns that have missing values in `test.csv` but had none in `train.csv`, (b) category values that never appeared in training, and (c) what your pipeline did with each. In 2–3 sentences, explain why this gap between training data and live data is one of the most common ways real ML systems fail.

12. Deliverables and submission

Submit one PDF on the leading2ai platform. The PDF must contain **both the questions and your answers**.

For every task, include:

1. the task heading and the task text (copy them from this document);
2. your code and its output: tables, numbers, charts;
3. your written answer, directly below the output it refers to.

How to make the PDF: in Jupyter, restart the kernel and run all cells from top to bottom, so the notebook runs cleanly in order. Then export it to HTML and print that to PDF from your browser, or export to PDF directly. Put the task text and written answers in Markdown cells.

- **File name:** Assignment2_FirstName_LastName.pdf
- **First page:** your name, the submission date and the dataset link.
- **Keep outputs short.** Print only what a task asks for. A 300-row printout wastes pages and hides your answer.
- **Charts must be legible in the PDF.** Check the exported file before you upload it.
- Do **not** include the raw data or the prepared CSV files in the PDF. Printing the final shapes and a few rows is enough.

Self-check before you upload:

- Restart and run all: the notebook runs top to bottom with no errors.
- Every task A1–F5 and every question G1–G8 has a visible answer.
- Every chart passes the §3 checklist, and each has a one-sentence takeaway below it.
- Your F2 checkpoint shows zero missing values in both train and test.
- Your data-preparation log covers every column you changed, dropped or created.

13. Grading rubric — 100 points

Part	Area	Points	What earns full marks
A	First look & quality (A1–A4)	12	Checks are complete, checkpoints match, and every flagged issue is backed by a value from the output.
B	Missing values (B1–B4)	14	"Absent" is correctly told apart from "missing" and proved with data. The MasVnrType loader issue is found and fixed. The LotFrontage investigation uses charts, not assertion.
C	Distributions & outliers (C1–C4)	14	Target skew is quantified and transformed. The two outlier rules are compared and their disagreement explained. Every outlier decision is justified and its effect is shown.
D	Relationships (D1–D4)	14	Charts are correctly ordered and readable. The heatmap is read critically. Redundant pairs are explained, not just listed.
E	Visualization & story (E1–E3)	12	The partial year and seasonality are noticed. The bad chart's distortion is quantified. The story has a clear context, insight and action aimed at the modeller.
F	ML-ready data (F1–F5)	18	The split comes before any learned statistic. Train and test have identical columns and zero missing values. The log is complete and honest. Up to 6 points are lost for any leakage from test into train.
G	Written questions (G1–G8)	16	Each answer is correct, in your own words, and tied to your own numbers or charts.

Across all parts, marks are deducted for charts that break the §3 checklist, for missing outputs, and for written answers that do not cite evidence.

14. Where each lecture section is assessed

Lecture section	Assessed by
EDA 1 · Inspecting a new dataset	A1, A2
EDA 2 · Data quality and inconsistencies	A3, A4, B2 (MasVnrType)
EDA 3 · Missing values	B1–B4, F2, G5
EDA 4 · Outliers and their impact	C3, C4, F2, G2
EDA 5 · Univariate and multivariate analysis	C1, C2, D1, D2
EDA 6 · Correlation	D3, D4, F3
EDA 7 · Feature–target relationships and modeling decisions	D2, D3, F1–F5, G1, G3, G4, G8
DataViz 1 · Choosing the right chart	Every chart, §3
DataViz 2 · Matplotlib (figures, subplots, labels)	C1, C2, E1, E2
DataViz 3 · Seaborn statistical plots	B3, D2, D3, D4
DataViz 4 · Distributions and relationships	B3, C1–C3, D3, D4
DataViz 5 · Time-series visualization	E1
DataViz 6 · Best practices and common mistakes	E2, §3 checklist, G7
DataViz 7 · Data storytelling	E3